

FLEA NEWS 30

Department of Entomology
Iowa State University
Ames, Iowa 50011 U.S.A.

June 1985

Compiled by Robert E. and Joanne H. Lewis

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Please find herewith:

- lists of additional literature for 1973 (list 16), 1974 (list 19), 1975 (list 18), 1977 (list 16), 1978 (list 17), 1979 (list 16), 1980 (list 12), 1981 (list 10), 1982 (list 7), 1983 (list 5), 1984 (list 3), 1985 (list 1)
- additions and changes to the mailing list
- notes on constructing a key to ceratophylline genera by F. G. A. M. Smit
- numbers for the species of fleas described in 1983

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G O O D - B Y E

Having finished the manuscript and illustrations for Volume VII (the last of the series, dealing with Malacopsyllidae and Rhopalopsyllidae) of the Rothschild Flea Catalogue, I am now saying a final farewell to the Magnificent Fleas - and to all magnificent colleagues. --- I shall certainly look back in gratitude.

Special thanks are due to my good friends, Bob & Joanne Lewis, for so successfully continuing to issue Flea News (I hope to remain on the mailing list in order to be kept informed of what is or will be going on in the Flea World).

It gives me pleasure to announce, with relief, that the Flea Collection in the British Museum (Natural History), London, has now been placed in the good care of Miss Annette Thomas. Please, do henceforth send reprints of your publications to her as well as requests, etc.

Frans G. A. M. Smit
7 June 1985

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HONORS

The Honorable Miriam Rothschild, CBE, DSc, FRS, HonFRES, HonFIB, has recently been recognized by a variety of august bodies in honor of her many accomplishments in the field of entomology. She has been

Elected a Fellow of the Royal Society;

Awarded an Honorary Doctorate degree by the University of Chicago;

Presented the Wigglesworth Medal by Her Majesty the Queen at the Royal Entomological Society of London Centenary;

Featured in a television special production, "A Usable Past", by the Smithsonian Institution on May 29;

Subject of a feature article by David McCullough entitled "A Rothschild who is known as the queen of the fleas - but at Ashton Wold, far from London, she also reigns over plants, dogs, owls and rabbits." Smithsonian 16(3): 139-153;

National Geographic Magazine is preparing an article featuring Dr. Rothschild to be published in the near future.

On behalf of its subscribers, the editors of Flea News extend our congratulations for these recent honors.

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ACCOLADES

Dr. Harry Hoogstraal, Director of Medical Zoology, NAMRU 3, Cairo, Egypt, was recently honored for his achievements by two American scientific societies. At the meetings of the American Society of Tropical Medicine and Hygiene, Dec. 2-6, 1984, at Baltimore, Maryland, the American Committee on Arthropod-Borne Viruses bestowed upon Dr. Hoogstraal The Richard Moreland Taylor Award for achievement in arthropod-borne virus studies.

The Entomological Society of America selected Dr. Hoogstraal to be the 1984 Founders' Memorial Award Lecturer at their national conference in San Antonio, Texas, Dec. 9-13, 1984. His lecture was entitled "Theobald Smith: His scientific work and impact." He also was elected Honorary Member of this Society.

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Professor LIU CHI-YING feted on the occasion of his 80th birthday - Recently a celebration was sponsored by the authorities of the Academy of Military Medical Sciences, Beijing, the Peoples' Republic of China, in honor of Professor Liu Chi-ying for his contributions to the fields of entomological research and teaching, on the occasion of his 80th birthday. Approximately 180 colleagues and friends attended and congratulatory speeches were delivered by Dr. Chu H.-f., President of the Entomological Society of China, and Prof. Kwan Z.-h., President of the Entomological Society of Beijing.

Professor Liu was born on 24 March 1905, in Mutu, near Lake Tai, west of Soochow, Kiangsu Province. He graduated from the University of Nanking in 1929. In 1933 he entered the graduate program of the Department of Entomology & Economic Zoology at the University of Minnesota, St. Paul, where he studied under Dr. William A. Riley, a medical entomologist and parasitologist, earning an M.S. degree. Upon returning to China he taught entomology and zoology, first at the National Chekian University and later at the National Kwansi University. Some years after the liberation he joined the Institute of Microbiology and Epidemiology, Academy of Military Medical Sciences, Beijing, where he resumed his researches in medical entomology, particularly in taxonomic studies on the Siphonaptera of China.

During his 50 some years of active service, Professor Liu has received many honors and has published in excess of 150 technical papers. Currently,

under the auspices of the Editorial Committee of Fauna Sinica, Academia Sinica, he is editing a Monograph of the Order Siphonaptera (of China) which is now in press.

The editors of Flea News join the many friends and colleagues of Professor Liu in congratulating him for his many years of service to his profession and in wishing him health, happiness, and many more years of productivity.

Information used here was provided in a brief biography of Professor Liu included in a letter from Professors Wang Dun-ying and Wu Hou-yong dated 17 April 1985.

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Skuratowicz Retires - In a recent letter from PROF. DR. WACŁAW SKURATOWICZ he informed us of his intention to retire on 1 September 1985. Prof. Skuratowicz passed his 70th birthday on 18 February 1985. Even though officially retired, he indicated his intention to continue his studies on the fleas of Poland in the Department of Systematic Zoology at the Adam Mickiewicz University, Poznań. We join the many friends and colleagues of Prof. Skuratowicz in wishing him a long, healthy, and productive retirement.

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On 7 June 1985 FRANS G. A. M. SMIT will celebrate his 65th birthday. His many friends and colleagues sincerely regret his departure from the flea world and wish to take this opportunity to wish him Happy Birthday and many more.

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NOTES ON THE CONSTRUCTION OF A KEY TO CERATOPHYLLINE GENERA *

* Key published in: R. Traub, M. Rothschild & J. Haddow, 1984, The Rothschild collection of fleas. The Ceratophyllidae. pp. 1-36, 205 figs.

Merely on the basis of the chaetotaxy of part of the outer side of the fore femur and of the inner side of the mid and hind coxa, Jordan (1933) divided Ceratophyllid genera of the Old and New World into three groups; the distinctions are no longer tenable. Ioff (1936), with limited material, divided some 20 genera/subgenera into six major groups ("Subtriben") within the family ("Tribus") and left three "aberrant" genera ungrouped.

Families of fleas are divided into subfamilies on the basis of only non-genitalic characters (such as cranial structure, frontal tubercle, eye, genal ctenidium, labial palp, antenna, pleural rod of mesothorax, pleural ridge of metathorax, striarium on metepimeron or basal sternum, chaetotaxy of coxa, tibia and distitarsomere, chaetotaxy of terga, tergal ctenidia and spinelets, ante-sensilial setae, sensillum and anal stylet of female). These non-genitalic structures are on the whole fairly uniform in Ceratophyllids and I have been unable to divide this large 'family' into satisfactory 'subfamilies'. Certain of the genera form reasonably neat seemingly natural groups but if one were to designate such groups as subfamilies, quite a large residue of ungroupable genera would be left, several of which deserving a subfamily status of their own. Stark (1970) remarked that "Apparently having undergone recent evolutionary radiation, this compact group [Ceratophyllidae] shows few phylogenetic gaps. Dividing the family and defining subfamilies and tribes is thus difficult, ..."

A few years ago I demoted this presumed family to subfamily status and placed Ceratophyllinae alongside Amphipsyllinae and Leptopsyllinae in Ceratophyllidae; Ischnopsyllidae and Ceratophyllidae together constituting the superfamily Ceratophylloidea (Smit, 1982, in: Synopsis and classification of living organisms. New York. pp. 557-563, pls. 121, 122).

In constructing a key to genera and subgenera I have tried to consider all morphological characters - not just a selection - of most of the species concerned. It was found that numerous structures are either too stable or too variable for generic usage. Tangible differences are not seldom elusive (occasionally by being too obvious to attract attention). As my tendency is for lumping, it was surprising to find that no fewer than 8 new genera and 9 new subgenera were erected during the study. Some genera are demoted to subgenera, no subgenera have been promoted to genera. Perhaps some more genera may eventually be treated as subgenera or vice versa, or certain amalgamations may take place later on. The fauna of fleas is still rather imperfectly known although the northern hemisphere, where most of the Ceratophyllids occur, compares favourably with the southern one.

Only about 800 hours (apart from some 300 hours used for drawing) could be spent on this particular task and the resulting key to genera is definitely not an entirely satisfactory one. As is usually the case, the more distinct taxa tend to come out first in the key, the more troublesome groups relegating themselves to the rear. It will be realised that most of the eighty or so different structures mentioned in the key will only show up in well-cleared and well-mounted specimens.

It is hoped that improvements to classification and key will gradually be made. A final word of advice: anyone considering an attempt (a) to divide the Ceratophyllinae into satisfactory 'natural groups', (b) to change generic or subgeneric definitions or (c) to compose a better key, must realise that this should only be done by a detailed study of all members of the family, not by selecting certain groups or geographical regions.

F. G. A. M. Smit

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RECENTLY PUBLISHED

The Fleas of Canada, Alaska and Greenland (Siphonaptera) by George P. Holland, 1985. (ISSN 0071-075X) Memoirs of the Entomological Society of Canada - No. 130, 631 pp., 508 figs., 109 maps. (Prices in \$Canadian 32.00 members, 41.00 non-members).

Workers familiar with the 1949 publication by this author on the Siphonaptera of Canada are already acquainted with the high standard of excellence that Dr. Holland has maintained throughout his professional career. His new book is further evidence, if more is needed.

As the author points out in the introduction, "..... we have been able to increase the total of Canadian, Alaskan and Greenland fleas to 183 valid forms from the 127 species and subspecies recorded from Canada in 1949. Furthermore, in the 36+ years that have elapsed there have been many changes in flea classification and nomenclature as well as increased information on distribution, host preferences, seasonal abundance and many other factors worthy of reporting. The end result of all this is embodied in the following monograph which undertakes to summarize all the taxonomic information available on the fleas known, or suspected, to occur within the political boundaries of

Canada, Alaska and Greenland, that is, all of the New World north of the southern border of Canada."

After the introductory notes and acknowledgments is a synoptic list of the taxa known from the area under consideration, followed by a tabular distribution by political areas. This is followed by a 15 page exposition on flea anatomy in relation to taxonomy, brief notes on the illustrations and a glossary of abbreviations used in the text and the keys. The next 451 pages constitute the systematic portion of the work, to which I shall return in the next paragraph. The book concludes with a host/flea index, a glossary of collectors, an extensive reference section, 109 distribution maps, and an index.

In his 1949 study, Holland (p. 53) states "From the standpoint of identification only, an artificial key, separating genera by means of more readily discernible characters, is far more satisfactory." [than keys to the families and subfamilies.] This is the approach followed in the present work, and the systematic section begins with such a key to the genera and subgenera. In the remainder of the section, however, the taxa are treated in conventional fashion, beginning with a short diagnosis of the order followed by treatments of the infraordinal categories of decreasing magnitude. Keys to the subfamilies, where recognized, are provided, as are keys to the species and subspecies where more than one taxon belonging to that genus is present in the area under consideration. Accounts for the genera, species, and subspecies all contain rather exhaustive synonymy sections with emphasis on the literature dealing with Canada, Alaska and Greenland. This is followed by a brief discussion of the taxon, and in the species and subspecies accounts, is concluded with a "New Records" section which, in some cases, may be rather lengthy. Illustrations are interwoven into the text in such a way as to make them readily accessible to the reader, a vast improvement over the custom of including all illustrations at the end of the text. Some of these were used in the 1949 volume but many have been redrawn or are new. In all cases, the quality of illustration is excellent.

The system of classification is that followed by most western students of the order. However, the 219 pages devoted to the Ceratophyllidae were too far advanced to permit conversion to the recent classification of this family as set down in Traub, Rothschild and Haddow (1983) and the older classification has been followed for this family. Since new classifications are seldom adopted quickly and, since all North American workers are more familiar with the older classification than they are with the new, I do not consider this to be a serious shortcoming. The synoptic list on pages 6 through 9 provides ample room for marginal updating should the reader wish to do so.

In summary, Dr. Holland has provided students of the order Siphonaptera with an invaluable tool in continuing studies of the fleas of North America. Much of the book applies to the fauna of the United States as well, and since we are currently lacking a comparable volume for this area, it will doubtless come to occupy a prominent position in North American workers' libraries. Congratulations, George, for a job well done! ---R.E.L.

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FLEA TRIVIA QUESTION: How many species of fleas are listed in the 10th edition of Systema Naturae?

Answer next time!

Dear Lord Rothschild. Birds, butterflies and history by Miriam Rothschild, 1983. (ISBN 0-86689-019-X) Balaban Publishers (Philadelphia/Rehovet) i-xx, 398 pp. + 12 colored plates and 92 black & white plates.

In spite of the fact that this book has been widely reviewed in the western countries, brief mention is made of it here for the benefit of colleagues in the east.

The central figure, as the title indicates, is Lionel Walter Rothschild (1868-1937), 2nd Baron Rothschild of Tring and uncle of the author. There are, however, other themes interwoven with the account of this eccentric and perplexing man. In short, it is a selective history of the British branch of the Rothschild family, with emphasis on Walter's contributions in the biological sciences and his role in precipitating the Balfour Declaration which ultimately led to the creation of the State of Israel.

Of perhaps most interest to recipients of Flea News is the inventory of natural history specimens amassed at the Rothschild Museum at Tring. The research collection contained 1400 mammal skins, 300,000 bird skins, 200,000 birds' eggs, 300 dried reptiles, 2½ million Lepidoptera including thousands of types, and a library of approximately 30,000 books.

While fleas do not figure prominently in this account, N. Charles Rothschild, the author's father, frequently collected for Walter, and it was he who collected and ultimately described the Plague flea, Xenopsylla cheopis. Photograph 58a in the grouping between pages 168-169 shows the camp at Shendi, Sudan, where this species was first discovered.

The book provides a wealth of history about the Rothschild family and a fascinating insight into the life and times of one of its more interesting members. ---R.E.L.

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George Henry Falkiner Nuttall and the Nuttall Tick Catalogue by James E. Keirans, 1984 (issued January, 1985). United States Department of Agriculture, Agricultural Research Service Miscellaneous Publication No. 1438, 1785 pp. Available from: Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402 USA. \$US 61.00. (Also available in microfiche from: National Technical Information Service, 5285 Port Royal Road, Springfield, Virginia 22161 USA.)

Since many of our readers work on other groups of ectoparasitic arthropods, brief mention of this compendium seems appropriate in the pages of Flea News.

This volume is exactly what the title states. It presents a biography of G. H. F. Nuttall and his 3-volume Tick Catalogue in its original and updated form. It also contains 4 appendices listing the type specimens of tick species, species names of Ixodoidea, host names, and localities mentioned in the catalogue.

There are also 5 non-tick records. These are:

Anoplura - N588, N3297, N3806
Hippoboscidae - N2460
Siphonaptera - N2337Z

This volume is a valuable addition to the literature dealing with ticks and Dr. Keirans is to be congratulated for his devotion to the project. However, the volume is inordinately large (according to my scales it weighs 11.5 pounds or 5.21 kilograms) and there is much wasted space. The use of condensed, or even elite type face, could have reduced it to 1/3 of its current size without inflicting any serious restrictions on its usefulness. ---R.E.L.

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American Insects. A Handbook of the Insects of America North of Mexico by Ross H. Arnett, Jr., Ph.D., 1985. (L.C.C.N. 84-15320, ISBN 0-442-20866-9) Van Nostrand Reinhold Company, Inc., New York, xiii, 850 pp., + many figs. (List price \$US 79.50, Institutional \$US 71.50). Available from: Flora & Fauna Publications, 2406 N.W. 47th Terrace, Gainesville, Florida 32606 USA.

While only 6 pages of this book are devoted to the order Siphonaptera (pp. 709-714), the entire tome should be of interest to at least North American entomologists, particularly those involved in teaching and research in insect systematics. This is not a replacement for a good General Entomology textbook, but it is certainly a worthwhile supplement.

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A REQUEST - Over the years a few colleagues have requested information concerning the locations of colonies of various species of fleas. Beyond the common species maintained by government laboratories we have, for the most part, been unable to provide assistance. In order to rectify this deficiency we would like to request that anyone knowing of the existence of cultures of any species of fleas pass this information on to us for inclusion in future issues of Flea News. Ideally, we would like to know the species involved, the location of the colonies and, if possible, the name of the person to contact for further information. Your cooperation in providing this information will be most appreciated.

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NOTICES

THE AMERICAN ASSOCIATION FOR ZOOLOGICAL NOMENCLATURE has recently been formed with the goal of building a support base in the United States that will provide direct financial assistance for the work of the International Commission on zoological Nomenclature through the International Trust for Zoological Nomenclature. They are seeking endorsement and support from various American scientific societies interested in zoological nomenclature in the hope of developing a long-term base of financial support for the International Trust from American scientists interested in nomenclature. Individual membership is also available. Additional information may be obtained from:

Dr. F. C. Thompson, Treasurer AAZN
U.S. National Museum of Natural History
Smithsonian Institution NHB-168
Washington, D.C. 20560 USA

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DR. LIU CHI-YING, Beijing, informs us that the subspecies Paradoxopsyllus intermedius humilangulus Hsieh, Yang & Li, cited in Liu C.-y., Wu H.-y., Liu Q. & Wu F.-l., 1982, Siphonaptera. Insects of Xizang, vol. II, p. 315, figs. 123, 124, is a mistake in typesetting and should be ignored. Since the name is now a part of the literature, it should be treated as a nomen nudum.

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CONFERENCE PAPERS

At the national meetings of the Entomological Society of America held at San Antonio, Texas, Dec. 9-13, 1984, one paper concerning fleas was presented:

"Field tests of Methoprene dust to control fleas on Mexican ground squirrels in south Texas" - J. T. Lang, T. J. O'Neill, W. Chamberlain & M. Francis

Scheduled for presentation at the annual meeting of the American Society of Mammalogists, Orono, Maine, 16-20 June 1985, is a paper entitled "Ectoparasite-mammal association patterns in the gallery forests of central Brazil." Presenter is Donald Gettinger, Dept. of Zoology, University of Oklahoma, Norman, Oklahoma 73109 USA.

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NEW JOURNAL - CLADISTICS, The International Journal of the Willi Hennig Society. At last the cladists have a forum of their own! CLADISTICS is published quarterly for the Willi Hennig Society by Meckler Publishing, 11 Ferry Lane West, Westport, Connecticut 06880, USA. Volume 1, number 1 (Winter, 1985) contains the following articles;

Brooks, D.R. & E.O. Wiley - Theories and methods in different approaches to phylogenetic systematics

Kluge, A.G. - Ontogeny and phylogenetics

Nelson, G. - Outgroups and ontogeny

Luckow, M. & R.A. Pimentel - An empirical comparison of numerical Wagner computer programs

Farris, J.S. - Distance data revisited

Platnick, N.I. - Philosophy and the transformation of cladistics revisited

Raven, R.J. - Symposium of cladistics, systematics and phylogeny: The first for Australia

The journal also publishes book reviews and notices of interest to Society members.

Subscriptions: Institutions \$95.00, individuals \$30.00 (Members includes \$5.00 membership fee); \$45.00 (non-members). Subscriptions outside the USA add \$5.00/year surface postage, \$20.00 air mail. Orders from North and South America should be directed to the Connecticut address; elsewhere to Meckler Publishing, 3 Henrietta Street, London WC2E 8LU, U.K.

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VIDEO FLEAS - A key to the fleas of medical and veterinary importance is available in VHS videotape. This 37-minute tape presents Dr. ALLEN H. BENTON in an explanation of some principles of flea taxonomy, together with video-thru-the-microscope details of fleas known to be pests of humans or domestic animals in the eastern USA. The cassette, together with the key in booklet form, may be ordered from BIOVIDEO, 3468 Route 83, Fredonia, New York 14063 USA, \$50 postpaid.

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MAILING LIST

ADDITIONS

- Ms. Tuti R. Hadi, National Institute of Health Research & Development, Ecology Research Centre, Jl. Percetakan Negara 29, P.O. Box 226, Jakarta, Indonesia
- Dr. K. E. Hyland, Department of Zoology, University of Rhode Island, Biological Science Center, B-114, Kingston, Rhode Island 02881 USA
- Wilton B. Kottkamp, Vector Control Office, DOCHMC, 801 S. Brentwood Road, Clayton, Missouri 63105 USA
- Dr. R. Lane, Medical Diptera Section, British Museum (Natural History), Cromwell Road, London SW7 5BD, England
- Lt Col J. T. Lang, USAF, BSC, Chief, Medical Entomology Function, USAF School of Aerospace Medicine (AFSC), Brooks Air Force Base, Texas 78235 USA
- Dr. G. S. Pfaffenberger, Department of Life Sciences, Eastern New Mexico University, Portales, New Mexico 88130 USA
- Dr. William H. Robinson, Department of Entomology, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061 USA
- R. E. Ryckman, Department of Microbiology, School of Medicine, Loma Linda University, Loma Linda, California 92350 USA
- Cdr. F. J. Santana, DPMIAC/AFPMB, Defense Pest Management Information Analysis Center, Forest Glen Section, WRAMC, Washington, D.C. 20307-5001 USA
- Miss Annette Thomas, Siphonaptera Section, British Museum (Natural History), Cromwell Road, London SW7 5BD, England
- Dr. Ye Rui-yu, Institute for Endemic Disease, Control and Research, Xinjiang Uighur Autonomous Region, China
- Dr. Yu Xin, Institute for Endemic Disease, Control and Research, Xinjiang Uighur Autonomous Region, China
- Barry J. Drucker, St. Louis County Health Department, 801 S. Brentwood Blvd., Clayton, Missouri 63105 USA

CHANGES

- Mr. Juan Carlos Morales, Department of Biology, Angelo State University, San Angelo, Texas 76909 USA
- Peter-Paul, 482-A Linden, San Francisco, California 94102 USA
- Dr. J. Ryba, Institut Hygieny a Epidemiologie, 100 42 Praha 10, Šrobárova 48, Czechoslovakia

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A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those species described in 1983. Readers are urged to call any omissions in these lists to our attention.

- 98301 sichuanna Wu, Chen & Zhai, 1983 Wagnerina
98302 heishuiensis Wang, Chen & Zhai, 1983 Amphalius spirataenius
98303 angustifrontalis Xie & Gong, 1983 Stenischia
98304 rupestis Xie & Gong, 1983 Stenischia
98305 montanis Xie & Gong, 1983 Stenischia
98306 humilis Xie & Gong, 1983 Stenischia
98307 eivissensis Beaucournu & Alcover, 1983 Rhadinopsylla (Rhadinopsylla)
98308 stirps Beaucournu & Orhan, 1983 Ctenophthalmus (Ctenophthalmus)
98309 quadrasetus Xie, Yang & Li, 1983 Ischnopsyllus (Hexactenopsylla)
98310 quintuesetus Xie, Yang & Li, 1983 Ischnopsyllus (Hexactenopsylla)
98311 magnabulga Xie, Yang & Li, 1983 Ischnopsyllus (Hexactenopsylla)
98312 tuoliensis Yu, Ye & Liu, 1983 Ophthalmopsylla (Ophthalmopsylla)
 volgensis
98313 postcurva Liu, Wu & Wu, 1983 Frontopsylla (Orfrontia) frontalis
98314 luski Schwan & Nelson, 1983 Catallagia

PROVISIONAL LISTS OF TAXA DESCRIBED IN 1984 AND 1985

1984

- Rhadinopsylla (Actenophthalmus) biconcava Chen, Ji & Wu
Nearctopsylla (Neochinopsylla) liupanshanensis Li, Wu & Liu
Lagaropsylla mytila Hårka
Parapsyllus magellanicus largificus Smit
Agastopsylla nylota euneomys Lewis
Hystrihopsylla weida qinlingensis Zhang, Wu & Liu
Palaeopsylla brevifrontata Zhang, Wu & Liu
Palaeopsylla anserocephoides Zhang, Wu & Liu
Geusibia minutiprominula minutiprominula Zhang & Liu
Geusibia minutiprominula ningshanensis Zhang & Liu
Geusibia longihilla Zhang & Liu
Callopsylla (Callopsylla) digitata Cai, Wu & Liu
Leptopsylla (Pectinoctenus) pectiniceps ventrisinulata Chen, Zhang & Liu
Ischnopsyllus (Hexactenopsylla) jinciensis Xiao

1985

- Peromyscopsylla himalaica australishaanxia Zhang & Liu
Peromyscopsylla scaliforma Zhang & Liu
Paradoxopsyllus calceiforma Zhang & Liu
Liuopsylla conica Zhang, Wu & Liu
Sigmatenus celebensis Lewis & Jones
Gryphopsylla hetera Lewis & Jones
Stivalius exoticus Lewis & Jones

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An asterisk (*) preceding a citation indicates we have seen neither the original publication nor verified it in another source. Anything appearing in brackets ([]) after a citation indicates an annotation, e.g., the name of a taxon described.

(1973 - 1980)

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1973 (list 16)

- Johnston, G. C. - The european rabbit flea in Tasmania. Tasman. J. Agr. 44(3): 157-158
- Parsons, R. E. - The host-flea relationships in a sylvatic plague endemic region of the south-western United States, with special studies on some new approaches to plague control. Diss. Abstr. Intl. B 34(1): 260

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1974 (list 19)

- *Salit, A. M., L. Sheta & G. M. Mourad - Ectoparasites on rodents in the Nile Valley - Upper Egypt. J. Egypt. Publ. Hlth. Assoc. 49: 3

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 18)

- *Mrčiak, M., L. Jedlička & I. Országh - Vývoj, súčasný stav e perspektívy medicínskej arachnoentomólgie na Slovensku. Biologica, Bratislava 30(2): 147-155

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 16)

- Hall, H. T. B. - Diseases and parasites of livestock in the Tropics. Longman Group Ltd., London, 288 pp. [Siph.: 236-237]
- McKenna, M. G., O. R. Larson & N. R. Fellows - Invertebrates of southwestern North Dakota: parasites. Inst. for Ecol. Studies, Univ. North Dakota, Res. Rep. 23: iii-xii, 1-97.
- *Ryba, J. - Blechy drobných savců a jejich hnízd v Tatrách. Zprávy ČSZS (10-12): 51

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1978 (list 17)

- *Ryba, J., M. Mrčiak & B. Rosický - Small mammals of the High Tatra Mts. as hosts of fleas /Siphonaptera/. II. - Congr. Ther. Inter. Brno 1978, Abstract of papers, p. 233

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 16)

- Berry, R. J. - The Outer Hebrides: Where genes and geography meet. Proc. R. Soc. Edinburgh 77B: 21-43 [Siph.: 28]
- *Stollman, A. & A. Dudich - Anotovaný prehľad drobných zemných cicavcov a ich ektoparazitov /Siphonaptera/ Klubinskej doliny a doliny Chmúra /Kysucké Beskydy/. Správy a informácie, Kysucké Múzeum 3: 109-121

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 12)

- *Abul Hab, J. - A list of arthropoda of medical and veterinary importance recorded from Iraq. Bull. biol. Res. Cent. (Baghdad) 12(1): 9-39

(1980 - 1981)

- *Rotshil'd, E. V. & I. S. Soldatkin - Contemporary views on plague enzooty and prognosis of natural foci during the economic exploitation of a territory. Vestn. Mosk. Gos. Univ., Geogr., Ser 5(4): 83-86

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 10)

- *Aikimbaev, M. A. & S. A. Aubakirov - Economic exploitation of a territory and its influence on the natural plague focus. In: "Prophylaxis of especially dangerous infections". Issl. protivochumn. Inst., Alma-Ata: 3-7
- *Aikimbaev, M. A., S. A. Aubakirov, O. S. Serzhanov & S. B. Pole - Epizootic state of territories enzootic for plague in the republics of Central Asia and Kazakhstan in 1976-1981. In: 11th Interrepublic Sci.-Pract. Conf. Plague Prophylaxis of Anti plague Insts. in Central Asia & Kazakhstan. Izd. Sredneaz. Nauch. Issl. protivochumn. Inst., Alma-Ata: 3-6
- *Cyprich, D., M. Kiefer & M. Krumpál - K problematike výskytu blých /Siphonaptera/ na vtákoch a ich hniezdach v podmienkach vel'kochovov. Celoštátna zool. konf. Bratislava 24-28.8.1981: 311-313
- *Cyprich D., M. Kiefer, M. Krumpál & J. Dúha - Charakteristika fauny blých /Siphonaptera/ pravobrežnej časti Bratislavy. Celoštátna zool. konf. Bratislava 24-28.8.1981: 277-279
- de Graaff, G. - The rodents of southern Africa. Butterworths, Durban/Pretoria, 267 pp. [Siph.: under species accounts]
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* * *

“And here’s the happy bounding flea—
You cannot tell the he from she.
The sexes look alike, you see;
But she can tell, and so can he.”

Roland Young
(1887-1953)
“The Flea,” *Not for Children*